Before Meaning, Before Choice

A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human–AI Return

From Readout Universe and Readout Genesis to a recursive human–world and Human–AI architecture

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Status K0, version 1.1 (source-lineage repair of the 6 September v1.0 after an independent review), 6 September 2026. The paper does not claim that Readout Universe or Readout Genesis is a completed law of nature, that the human-domain quotient has been empirically closed, or that the equations below are literal neuroscience. It advances a narrower but strong contribution: a typed derivational architecture in which human-semantic and agentic categories are admitted only after retained difference, reader-conditioned access, domain translation, explicit defect/loss accounting, and claim-status control. It makes no global priority claim: the reviewed neighbouring literatures contain close ancestors for most local components, and the contribution is stated as same / different / cited against each of them (Table 1). Every internal source is cited by a Zenodo DOI or by a commit-pinned public git blob (Section 21).

Central claim. Human experience and choice should not be inserted as primitive objects at the beginning of a theory. Under a readout-native architecture, a retained difference must first become structurally accessible under a declared reader and domain translation. Meaning is then a domain-level significance relation, experience is a lived configuration of encountered trace and meaning, retained experience can reshape later accessibility, and overt choice is selected only from a history-shaped field of live possibilities. Human–AI interaction is a special mediated loop inside this architecture: the prompt is a bounded articulation of an already history-bearing human state, AI output begins as a knowledge-like candidate rather than certified knowledge, and any claimed human gain must survive a session boundary and an unaided Human Return test. The resulting programme is therefore not *meaning as primitive* $\rightarrow$ *choice*, but *retained difference* $\rightarrow$ *admissible readout* $\rightarrow$ *meaning* $\rightarrow$ *experience* $\rightarrow$ *live possibility* $\rightarrow$ *choice* $\rightarrow$ *action* $\rightarrow$ *new retained difference*.

Abstract

The Human–AI Readout Programme has developed a sequence of papers on human potential, learning, retained experience, semantic mobility, Human–AI coupling, meaning-giving, live possibility, effective agency, and Human Return. Read separately, these works can appear to be a family of related conceptual models. This paper argues that they can be reconstructed as a single derivational architecture only if the chain is anchored one level deeper, in Readout Universe and Readout Genesis. The root contains no built-in final labels such as mind, meaning, experience, memory, agency, choice, or AI. It begins with retained distinctions, ordered history, reader capability, finite state transition, and a requirement that any domain enter through a sufficient, dynamically coherent translation with explicit defects and bounded claims. The paper then derives a human-domain river in stages. Readout Genesis and its MEMK domain supply the separation of world occurrence, retained record, event trace, meaning, experience, memory, and knowledge. Human LoRA supplies selective cross-time persistence. From Problem to Hypothesis supplies graded and history-shaped accessibility. Choice Begins Before Choice inserts a Live Possibility Field between feasible capacity and overt choice. Potential as a Readout supplies the read–decide/refuse–execute–feedback envelope over a witnessed option set. Fusion/Tunnel inserts Human–AI dialogue as a recursively history-sensitive mediation, while CTSA Human Return locates the measurement boundary after AI removal. The paper places this architecture under pressure from mature neighbouring literatures: Batesonian difference, computational mechanics, the information bottleneck, bisimulation and MDP homomorphisms, phenomenology, enactivism, ecological affordances, the capability approach, temporal theories of agency, epistemic injustice, theories of power and structural violence, active inference, and empirical Human–AI decision research. The conclusion is deliberately asymmetric: most local phenomena are already established elsewhere; the candidate contribution lies in the derivational discipline that keeps root structure, domain meaning, lived experience, practical possibility, overt choice, evaluator-visible behaviour, epistemic warrant, and durable Human Return from collapsing into one another. The paper ends with a canonical Root-to-Human Master Equation River, a claim-status ledger, rival-model tests, and explicit removal conditions. If the readout-native layers add no stable explanatory or predictive value beyond simpler alternatives, they should be compressed or removed.

Keywords: Readout Universe; Readout Genesis; retained difference; meaning; experience; live possibility; agency; Human–AI interaction; Human Return; epistemic architecture; domain translation; corrigibility.

1 The Problem Is Not Another Theory of Choice

The immediate temptation is to begin with a familiar object: a person has preferences, represents options, deliberates, chooses, acts, and receives feedback. A second temptation is to move one step earlier and ask how experience, embodiment, social norms, prior learning, or language shape the options that enter deliberation. A third temptation is to move still earlier and treat meaning or sense-making as the primitive from which action is organised. Each move captures something important. None, however, answers the methodological question that motivates the present paper: what licenses us to place "meaning", "experience", "option", "choice", or "agency" into the ontology in the first place?

Readout Universe and Readout Genesis were designed precisely to block premature naming. Their information-philosophy constitution places difference before entity, ordered history before container-time, retention before persistence semantics, reader-conditioned access before domain truth, and domain translation before domain-final nouns [Lahtee, 2026l, Readout Genesis Programme, 2026]. This discipline changes the order of explanation. Human cognition is not denied; it is delayed until the architecture has earned the right to use cognitive vocabulary.

The programme's later papers can then be reread as successive answers to a single question. When a retained difference becomes readable to a human, how does it become significant, lived, remembered, practically accessible, chosen, enacted, corrected, and eventually returned as human capability after AI mediation? The Master Equation River audit showed that the internal corpus already contains most of these transitions, but also identified notation collisions and a deeper missing anchor: the river began too late, at the human readout layer, rather than at the non-semantic root [Lahtee, 2026g].

This paper performs that anchoring. It does not claim that every human behaviour proceeds through explicit deliberation, that reflexes require a live possibility field, or that all cognition is linguistic. The primary target is practical human agency: situations in which a route can count as a candidate for acceptance, refusal, negotiation, delay, delegation, exit, revision, inquiry, or other purposive action. The architecture is therefore broader than verbal deliberation but narrower than a complete theory of biological behaviour.

2 What the Literature Already Owns

A defensible contribution claim must begin by conceding the strength of neighbouring literatures. The present architecture is not the earliest to treat difference, representation, perspective, meaning, affordance, agency, temporality, power, or Human–AI influence as serious theoretical objects.

2.1 Difference, predictive equivalence, and compressed representation

Bateson's well-known formulation of information as a difference that makes a difference is a clear ancestor of any difference-first programme [Bateson, 1972]. Computational mechanics goes further by constructing causal states as equivalence classes of histories that are predictively indistinguishable, providing a principled way to identify structure relative to future prediction [Shalizi and Crutchfield, 2001]. The information bottleneck similarly formalises representation as selective compression that preserves information relevant to a declared target [Tishby et al., 1999]. The present work therefore claims no priority for "difference", history-based state construction, reader-relative equivalence, or relevance-sensitive compression.

2.2 Structure-preserving abstraction

The Readout Genesis domain-weld condition has close mathematical relatives in lumpability, bisimulation, state abstraction, and MDP/SMDP homomorphisms. These traditions ask when a lower-dimensional or quotient representation preserves transition- or value-relevant structure [Ravindran and Barto, 2003, Ferns et al., 2011]. Whatever the domain weld contributes, it cannot be the bare fact that structure-preserving abstraction exists. The residual move is epistemic: a dynamics-preserving translation is used as a gate before final semantic categories are admitted, and every approximate bridge must expose what it loses.

2.3 Meaning, experience, and sense-making

Phenomenology has long treated experience as intentional and horizon-laden; enactivism treats cognition as embodied sense-making arising in organism–environment coupling rather than passive extraction of pre-packaged meaning [Varela et al., 1991, Di Paolo, 2005, Thompson, 2007]. The programme's claim that meaning is relational, embodied, history-conditioned, and action-relevant therefore has major predecessors. The paper does not claim to discover sense-making. Its question is instead how such significance can be placed in a typed architecture after a non-semantic retained-state layer and then carried forward without collapsing meaning into truth, experience into memory, or retention into improvement.

2.4 Action possibilities, real freedom, and the conditions of choice

Ecological psychology's affordance tradition already treats possibilities for action as relational between organism and environment [Gibson, 1979]. Sen's capability approach distinguishes formal resources from real opportunities; Kabeer links resources, agency, and achievements while emphasising the conditions under which meaningful choices are made [Sen, 1985, Kabeer, 1999]. Emirbayer and Mische treat agency as temporally embedded rather than a timeless instant [Emirbayer and Mische, 1998]. The Live Possibility Field therefore cannot claim that options are agent-relative or that context matters. Its narrower residual proposal is that feasible action and currently live practical consideration should be separated as distinct readout objects.

2.5 Power, interpretive resources, and structural deprivation

Foucault's account of power acting upon possible action, Pettit's non-domination, Fricker's hermeneutical injustice, and Galtung's structural-violence framework all undermine the idea that freedom can be read directly from visible non-interference or observed behaviour [Foucault, 1982, Pettit, 1997, Fricker, 2007, Galtung, 1969]. The present framework does not replace these traditions. It supplies a narrower measurement interface: a structural mechanism may affect agency by changing the accessibility distribution over live routes before an overt choice occurs. Calling that change domination, injustice, or violence still requires an independent normative argument.

2.6 Integrated agent models and Human–AI coupling

Active inference is the strongest formal neighbouring architecture because it integrates perception, belief updating, action, learning, and policy selection within a single framework [Friston et al., 2017]. Human–AI decision research likewise documents path dependence, reliance, explanation effects, and the need for common evaluation frameworks [Lai et al., 2023]. Glickman and Sharot provide direct evidence that repeated interaction with biased AI can alter later human perceptual, emotional, and social judgements, demonstrating that the human state entering a later interaction need not be fixed [Glickman and Sharot, 2025]. The present programme therefore does not claim that history changes future judgement or that AI can influence users. Its residual contribution is the explicit chain from pre-prompt human state to candidate status, independent resistance, selective human retention, and unaided Human Return.

3 THE READOUT CONSTITUTION: BEFORE MEANING

Readout Universe is not used here as decorative metaphysics. Its information philosophy is operational: the root must not contain domain-final nouns; difference precedes the name of what differs; ordered history precedes time treated as a container; retention precedes persistence semantics; readers create access rather than retroactive root truth; domains are translations; meaning comes after readout; non-exact translation must expose loss; claims cannot borrow certainty across layers; and reporting is the last stage [Lahtee, 2026l,m]. The whitepaper states the governing rule in one line: "Information philosophy fixes the order by which meaning becomes admissible and governs every translation" [Lahtee, 2026l, l. 35].

The most compressed form is

$$
\begin{aligned}
&\text{Retention} \to \text{Structure} \to \text{Translation}\\
&\to \text{Readout} \to \text{Meaning} \to \text{Report},
\end{aligned}
\tag{1}
$$

with the stronger MEMK extension (the MEMK specification's own mandatory order reads Retention → Structure → Translation → Meaning → Report [Readout Genesis Programme, 2026, l. 14]):

$$
\begin{aligned}
&\text{Retention} \to \text{Structure} \to \text{Candidate State}\\
&\to \text{Sufficiency} \to \text{Quotient} \to \text{Domain Dynamics}\\
&\to \text{Readout} \to \text{Meaning} \to \text{Experience}\\
&\to \text{Memory} \to \text{Knowledge} \to \text{Checked Report}.
\end{aligned}
\tag{2}
$$

Two consequences are load-bearing. First, semantic categories are not root primitives; the MEMK file lists "meaning is not a root primitive" among its root non-claims [Readout Genesis Programme, 2026, l. 50]. Second, formal success at one layer cannot certify empirical success at another. A commuting quotient would not prove that a psychological interpretation is true; a stable human report would not prove that the report exhausts the world occurrence.

3.1 Root state and finite stepper

In the Genesis specification the non-semantic retained state is written

$$S_n = (G_n, \Lambda_n, T_n), \tag{3}$$

where G_n is a retained relational carrier, Λ_n retained labels or typed distinctions, and T_n an append-only tape of lineage, residues, and deviations. The root transition interface is

$$S_{n+1} = F(S_n, u_n, c_n, T_n). \tag{4}$$

The critical separation is

$$S_n \neq Z_{\text{MEMK},n} \neq D_{\text{MEMK},n}. \tag{5}$$

Retained root state, candidate semantic–cognitive representation, and discovered domain quotient are different objects; the MEMK candidate state $Z_{\text{MEMK},n}$ is written out as an explicit tuple in the public v0.2.0 file [Readout Genesis Programme, 2026, l. 80]. This is the earliest point at which the present paper differs from theories that begin with beliefs, preferences, policies, conscious states, or semantic representations. The claim is not that such states are unreal. It is that their use is downstream of an admission procedure.

4 DOMAIN WELD: HOW HUMAN VOCABULARY IS ADMITTED

A domain is not created by attaching a familiar noun to the root. Readout Genesis requires a translation q_D whose domain dynamics preserve the relevant root dynamics and readouts. Schematically,

$$q_{D,n+1} \circ F_n = F^{\sharp}_{D,n} \circ q_{D,n}, \tag{6}$$

with corresponding readout and invariant conditions. This is structurally analogous to familiar abstraction and homomorphism ideas, but here the weld is epistemic as well as mathematical: it governs whether a domain-level statement may be emitted at all.

For human-semantic work, exact closure has not been established. The correct status is therefore approximate and defect-accounting. Let $\tilde{q}_H$ be a candidate human-domain translation. A valid implementation must declare a defect vector

$$\varepsilon_H = \mathrm{Def}\big(\tilde{q}_H \circ F,\ F_H^{\sharp} \circ \tilde{q}_H,\ \mathscr{O}_H,\ \mathrm{Inv}_H\big), \tag{7}$$

and a preregistered acceptance rule $\|\varepsilon_H\| \le \tau_H$ if a quantitative gate is attempted. Without a declared metric and tolerance, the status remains unresolved rather than silently "close enough".

This defect discipline is more important than the exact formula. It means that every human-level construct in the rest of the paper is typed by its origin and by what the translation may have discarded. The architecture therefore resists a common philosophical shortcut: observing that a high-level concept is useful and then treating that usefulness as evidence that the concept was present in the root all along.

5 From Retained Record to Meaning and Experience

MEMK supplies a crucial distinction between event occurrence, retained event record, and domain-accessible event trace. Let A_n denote the occurrence under consideration, r_n the retained record before semantic naming, and x_n the MEMK-accessible trace. Then

$$A_n \neq r_n \neq x_n. \tag{8}$$

This need not imply that every pair is always distinguishable under every reader; it states that the roles must not be collapsed by default [Readout Genesis Programme, 2026].

A finite human readout of the accessible trace is represented schematically as

$$\rho_n^H = R_H(x_n \mid H_n, c_n, Q_n), \tag{9}$$

where H_n summarises the current human-domain state, c_n context, and Q_n an active question or criterion. Meaning is then a significance relation, not a substance carried inside x_n:

$$\mu_n = \Psi_H(\rho_n^H, H_n, c_n, Q_n). \tag{10}$$

The strong-form experience paper separates several analytic modes,

$$\mu_n = (\mu_n^{\mathrm{aff}}, \mu_n^{\mathrm{prag}}, \mu_n^{\mathrm{auto}}, \mu_n^{\mathrm{conc}}, \mu_n^{\mathrm{epi}}), \tag{11}$$

without treating them as biological modules [Lahtee, 2026c]. Experience is then

$$E_n = \Phi_E(x_n, \mu_n, \gamma_n^{\mu}, c_n), \tag{12}$$

where γ_n^{μ} denotes the effective strength or control of active meaning over the current experience (the MEMK file writes this argument as κ_n, Readout Genesis Programme, 2026, l. 92; it is renamed here to avoid a collision with semantic accessibility, following the Master River audit [Lahtee, 2026g]; the same equation appears as eq. 14 of the deposited Genesis synthesis, Lahtee, 2026k). The conceptual compression is

$$\text{Experience} = \text{phenomenon-as-meaningfully-read}. \tag{13}$$

This is not a claim that the reader makes the phenomenon true. It is a claim about the level of lived significance. Meaning-giving can be false, partial, manipulative, culturally inherited, or later revised.

5.1 Naming comes later, but can loop back

Meaning Before Naming allows lived significance to precede stable lexical articulation:

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n), \qquad E_n, \mu_n \text{ may precede stable } \ell_n. \tag{14}$$

But the later Experience paper corrects any one-way reading. Naming can reorganise what it articulates:

$$\mu_n \to E_n \to \ell_n \to \mu_{n+1} \to E_{n+1}. \tag{15}$$

Thus "meaning before naming" is a temporal possibility, not a doctrine that language is irrelevant or merely epiphenomenal [Lahtee, 2026h,c].

6 Retention: The Reader Can Change

Experience is not automatically learning. Human LoRA supplies the distinction between exposure, retained residue, and selective operator change. In a generic dependency form,

$$H_{n+1} = U_H\big(H_n,\ \mathrm{Retain}(E_n, \mu_n, \rho_n^H),\ \delta_{n:n+1}^{\mathrm{world}},\ X_{n:n+1}^{\mathrm{other}}\big). \tag{16}$$

Only selected residues are carried forward; world correction and other experience remain co-determining inputs [Lahtee, 2026f,d]. This yields a central temporal asymmetry:

$$\mathrm{Retain}(E_n) > 0 \implies H_{n+1} \neq H_n \quad \text{possible, not guaranteed.} \tag{17}$$

The same biological individual may therefore confront the next situation with a different effective interpretive operator. This makes the entire downstream theory of possibility history-sensitive before any new choice is made.

7 History-Shaped Readout-Accessibility

The semantic-mobility paper supplies the formal bridge from retained history to graded access. Reachability is not accessibility. For a semantic route $e : s \to s'$, let $\kappa_{A,t,Q}^{\mathrm{sem}}(s' \mid s)$ be a finite access weight. Then an alternative may be structurally reachable while still having low practical access [Lahtee, 2026j].

Table 1: Contribution boundary. Local mechanisms have mature ancestors; the residual contribution is the typed derivational interface. Comparison vocabulary: same / different / cited.

Component	Strong neighbours	Residual role in this paper
Difference / retained history	Bateson; computational mechanics	Not claimed as this paper's; retained difference is used as a non-semantic root object that precedes domain naming.
Reader-relative compression / equivalence	Information bottleneck; predictive equivalence	Require declared reader, provenance, defect/loss, and bounded claim status; equivalence is never promoted to absolute identity.
Structure-preserving abstraction	Lumpability; bisimulation; MDP/SMDP homomorphism	Use dynamic preservation as an epistemic admission gate for semantic domains rather than as state reduction alone.
Meaning / sense-making	Phenomenology; enactivism	Derive meaning only after readout and translation; keep meaning distinct from truth and warrant.
Action possibility	Affordances; capability approach	Insert a graded live possibility field between feasible capacity and overt choice.
Temporally embedded agency	Emirbayer–Mische; learning and memory literatures	Make accessibility explicitly history-shaped and allow the effective reader at $t+1$ to differ from the reader at t.
Power before overt choice	Foucault; Pettit; Fricker; Galtung	Provide a readout/live-field interface without equating all social shaping with domination or violence.
Integrated perception–action dynamics	Active inference	Begin one level earlier with a non-semantic root and require domain translation, loss accounting, and claim-tier separation.
Human–AI feedback	Human–AI decision literature; Glickman–Sharot	Connect feedback to pre-prompt state, candidate status, resistance, retention direction, and unaided Human Return.

Two history descriptors are kept separate. Semantic attraction is destination-sensitive,

$$a_{Q,t}(e) = \left[\Phi_{Q,t}(s) - \Phi_{Q,t}(s')\right]_+, \tag{18}$$

where $\Phi_{Q,t}$ is a declared access landscape, not a physical potential. Recent-path momentum is path-sensitive,

$$m_{t+1}(e) = \rho\, m_t(e) + \mathbb{1}[e_t = e], \qquad 0 \le \rho < 1. \tag{19}$$

A candidate history-shaped access law is

$$\begin{aligned}\kappa^{\text{sem}}_{t+1}(e \mid Q) \;\propto\; &\kappa^{(0)}_t(e \mid Q) \\ &\times \exp\left[\beta a_{Q,t}(e) + \mu_m m_t(e) - \nu c_{A,t,Q}(e) + \xi_t(e)\right].\end{aligned} \tag{20}$$

This equation remains OPEN: it is a typed finite hypothesis, not a validated universal cognitive law. Its theoretical role is to prevent structural availability, attraction, recent traversal, and control cost from collapsing into one access variable.

Rhythm and resonance are optional modifiers rather than root laws. Rhythm is a domain readout of structured recurrence on ordered lineage; resonance is a domain diagnostic of congruence between a current externally prompted experience and retained experiential organisation [Lahtee, 2026c]. Neither is required for every case of live possibility formation. If they add no incremental explanatory value beyond simpler temporal, schema, familiarity, or retrieval variables, they should be removed from the technical model.

8 BEFORE CHOICE: THE LIVE POSSIBILITY FIELD

The central transition from experience theory to agency theory is the insertion of a live possibility layer. Let $\Pi^{\text{phys}}_t(g)$ denote physically possible policies for task g, and $\Pi^{\text{feas}}_{A,t}(g)$ those policies the agent could access, use, or learn within declared resources, time, and structural conditions. The current live set is narrower:

$$\Pi^{\text{live}}_{A,t}(g) \subseteq \Pi^{\text{feas}}_{A,t}(g) \subseteq \Pi^{\text{phys}}_t(g). \tag{21}$$

A practical accessibility profile is

$$L_{A,t}(g) = \left\{\left(\pi, \lambda^{\text{live}}_{A,t}(\pi \mid g)\right) :\ \pi \in \Pi^{\text{feas}}_{A,t}(g)\right\}, \tag{22}$$

with an operational, study-specific threshold

$$\Pi^{\text{live}}_{A,t}(g) = \left\{\pi :\ \lambda^{\text{live}}_{A,t}(\pi \mid g) \ge \tau_{\text{live}}\right\}. \tag{23}$$

The threshold is a measurement device, not a universal psychological constant [Lahtee, 2026a]. Overt choice is downstream:

$$\pi^{\text{choice}}_{A,t} \in \Pi^{\text{live}}_{A,t}(g). \tag{24}$$

Enactment may diverge from the chosen policy, and evaluator-visible records need not reveal the underlying trajectory:

$$\begin{aligned}&\pi^{\text{act}}_{A,t} \neq \pi^{\text{choice}}_{A,t} \text{ is possible,} \\ &Y_{\text{obs}} = \mathscr{O}_q(H_{0:T}), \qquad Y_{\text{obs}} \neq H_{0:T}.\end{aligned} \tag{25}$$

The non-collapse chain is therefore

$$\begin{aligned}&\text{possible} \neq \text{feasible} \neq \text{live} \\ &\neq \text{chosen} \neq \text{enacted} \neq \text{observed}.\end{aligned} \tag{26}$$

The claim "choice begins before choice" is now sharpened. It does not mean that a retained difference is itself a choice. It means that the architecture of a deliberative choice has ancestral conditions in what was retained, what a reader can discriminate, which domain translation makes a route intelligible, and which history-shaped accessibility profile allows that route to enter practical consideration.

9 EFFECTIVE, CORRIGIBLE AGENCY

The Live Possibility Field does not replace the agency-potential model. It inserts a pre-choice restriction inside the agent's envelope. For task g, the agency architecture requires four jointly relevant conditions: Read_g for reading relevant distinctions, D_g for actor-directed decision or refusal, X_g for causal enactment, and F_g for feedback, objection, and revision. In its current form the task-specific potential is read over the *witnessed* option set:

$$p^*_{A,g}(h,z;T,B,P) = \max_{\pi\in\Pi^{\mathrm{wit}}_A(g;h,z,T,B)} \Pr^{\pi}_P\left(\mathrm{Read}_g \cap D_g \cap X_g \cap F_g\right), \tag{27}$$

where Π^{wit}_A is the finite set of policies for which a retained record exists that an agent comparable to A under a declared comparability relation (shared declared fields of z and a history window of declared length) executed the policy within horizon T and budget B [Lahtee, 2026i, eq. 1]. A merely conceivable, or merely permitted, policy is not in the set; the maximum is therefore computable from records, and the unread remainder is a non-readout rather than a smaller number waiting to be found. The v1.0 draft of this paper wrote the maximum over $\Pi^{\mathrm{feas}}_A(g)$, following the programme note that *Potential as a Readout* has since superseded; the four events are unchanged, the domain of the maximum is not, and the later paper adds a second layer ($p^{*(2)}_{A,g} = \max_{z\in J_{\mathrm{feas}}} p^*_{A,g}$, the recoverable gap being the difference of layers) and two non-collapse rules: potential $\neq$ exercised $\neq$ observed, and diagnosis of compression $\neq$ attribution of responsibility [Lahtee, 2026i,a].

This definition explains why potential, performance, and observation can diverge. A person may possess a feasible route that is not currently live; may make a choice that cannot be enacted; may act under third-party control; or may preserve objection and revision capacity that an evaluator fails to observe. Conversely, a visible successful outcome does not prove actor ownership, corrigibility, or durable capability.

9.1 Power before prohibition

Once live accessibility is explicit, a narrower political thesis follows:

$$\text{Power may alter agency by altering meaning/accessibility before overt choice.} \tag{28}$$

This is not a claim that all socialisation is domination. Human beings necessarily inherit languages, norms, roles, skills, and interpretive repertoires. The architectural point is that social and institutional conditions can change whether refusal, appeal, delay, exit, negotiation, or alternative interpretation becomes practically live before any direct prohibition is imposed.

A narrowed live field is not by itself a violence score. A structural-deprivation claim requires at minimum: a specified agency-relevant loss, a feasible counterfactual, an independent normative reason why the blocked route constitutes harm, domination, injustice, or violence, and, if responsibility is to be attributed, a chooser of the regime that produced the closure [Galtung, 1969, Lahtee, 2026a,i]. Diagnosing the closure and attributing it are separate acts.

10 HUMAN–AI MEDIATION BEGINS BEFORE THE PROMPT

The root-to-human derivation changes how Human–AI interaction is conceptualised. A prompt is not the whole human state. It is a bounded articulation:

$$H_t \xrightarrow{L_H} Q_t, \qquad Q_t \neq H_t. \tag{29}$$

Because H_t is itself a domain-level product of retained history and prior readout, the prompt is a readout of a readout, not an unconditioned starting point. The AI receives Q_t and returns Y_t:

$$Q_t \to \mathrm{AI}_t \to Y_t. \tag{30}$$

Under the candidate contract,

$$\mathrm{AI}(Q_t) = K_{\mathrm{like}}, \qquad K_{\mathrm{like}} \neq K_{\mathrm{validated}}. \tag{31}$$

Fluency, citations, or formal appearance do not upgrade status by themselves [Lahtee, 2026d]. The AI output then re-enters the human domain as another mediated encounter:

$$Y_t \xrightarrow{R_H} E^{\mathrm{AI}}_{H,t}. \tag{32}$$

If a residue is retained,

$$H_{t+1} = U_H\left(H_t, E^{\mathrm{AI}}_{H,t}, \delta^{\mathrm{world}}, X^{\mathrm{other}}\right), \tag{33}$$

and therefore

$$L_{A,t+1} \neq L_{A,t} \tag{34}$$

may hold before the next prompt is produced. Glickman and Sharot's experiments provide direct external evidence for the general history-sensitivity claim: repeated interaction with biased AI altered later human judgements across multiple domains [Glickman and Sharot, 2025]. They do not validate the full readout architecture, but they defeat the assumption that every prompt begins from a fixed human baseline.

10.1 Difference, resistance, and retained human agency

Fusion/Tunnel defines durable human epistemic expansion such that three conditions are non-substitutable: Difference, Resistance, and retained human Agency [Lahtee, 2026d]. Under that stipulated target,

$$D > 0, \qquad \mathrm{Resist} > 0, \qquad A_H > 0 \tag{35}$$

are constitutive necessities, not empirical frequency laws. No new readout-distinguishable route means no expansion beyond the retained frame; no independent resistance means no principled separation between correction and self-confirming proliferation; no retained human agency means possible system gain without demonstrated human-potential gain.

Within-session reciprocal amplification is epistemically neutral. A dialogue can multiply defeaters or multiply confirmation. The Master River audit therefore proposes, as programme notation, a separation of candidate update from retention gate:

$$\widetilde{\Delta\Omega}_s^H = B_s A_s, \qquad \mathrm{rank}(B_s A_s) \ll d_H, \tag{36}$$

$$\Omega_{s+1,0}^H = \Omega_{s,0}^H + \eta_s \widetilde{\Delta\Omega}_s^H, \qquad 0 \le \eta_s \le 1. \tag{37}$$

This is a functional low-rank analogy, not a claim that brains literally implement engineering LoRA. Two source facts are recorded so that the lineage is exact: the deposited Fusion/Tunnel v8.1 keeps η_s only as a conceptual session-end gate and estimates durable residue through an unaided return battery $Y_{s+\Delta}^{\mathrm{return}} = (R_{\mathrm{rec}}, R_{\mathrm{disc}}, T_{\mathrm{new}}, Q_{\mathrm{next}})$ and a difference-in-differences estimand RET (its Repair 7), and its three-axis outcome $J_s^* = (\mathrm{AUG}_s, \mathrm{SYN}_s, \mathrm{RET}_s)$ is not to be collapsed into one score [Lahtee, 2026d]; equations (36)–(37) are the audit's proposal for writing the same intent without a doubled gate [Lahtee, 2026g].

11 Human Return: The Upper Measurement Boundary

The programme's strongest practical boundary is not the quality of the coupled Human–AI answer. It is what remains demonstrably usable by the human after AI support is removed. CTSA classifies retained content/access as Conceptual Return, Tool-Selection Return, Skill-Execution Return, or Alternative-Generation Return [Lahtee, 2026b]. A compact Human Return record is

$$H_{\mathrm{return}} = \langle G_{\mathrm{CTSA}}, L, M, P, W, \Delta_{\mathrm{dir}} \rangle, \tag{38}$$

where G_{CTSA} is demonstrated gain, L loss, M metacognitive regulation, P provenance, W warrant/effective resistance, and Δ_{dir} epistemic direction (CTSA itself writes this record as R_H; the symbol is renamed to keep R free for read, resistance, resonance and rhythm). The tuple is bookkeeping architecture, not a universal scalar. The corresponding non-collapse laws are

$$\text{Exposure} \neq \text{Retention} \neq \text{Improvement}, \tag{39}$$

$$\text{Assisted performance} \neq \text{Unaided Human Return}. \tag{40}$$

A persistent misconception is still persistent; CTSA records it without granting knowledge status. A user may gain a skill while losing uncertainty sensitivity, or gain alternatives while becoming less able to distinguish which ones are warranted. Gross gain therefore cannot define net human expansion.

12 The Canonical Root-to-Human Master Equation River

The programme can now be written as one typed recursive river (Figure 1). The arrows denote admissible dependency or transition structure, not one universal causal law. Each segment retains its own claim status.

The deepest consequence is that the cycle does not return to the same reader by default. When retained change is non-zero, the next encounter is read under a modified human state. Experience can therefore change the reader; the changed reader can change what becomes live; what becomes live can change choice; choice changes action and the world; the changed world supplies new returned records; and the next readout begins from a different lineage.

13 Provenance and Claim Status Are Part of the Equation

A major difference between a compact theory diagram and a readout-native architecture is that the arrows themselves carry provenance and status. Let each transition e_i in the river have a ledger

$$\Lambda(e_i) = \langle \mathrm{Prov}_i,\ \mathrm{Tier}_i,\ \mathrm{Def}_i,\ \mathrm{Reader}_i,\ \mathrm{Falsifier}_i \rangle. \tag{41}$$

The river is not fully specified without this ledger. This is the methodological signature of the paper. A strong phrase in prose is not automatically a strong empirical claim. The architecture permits rhetorical compression at the title level while forcing technical status to remain visible underneath.

14 Why This Is Not Simply Active Inference

The most serious formal objection is that perception, learning, action, and history can already be modelled under active inference. The objection is partly correct. Active inference is more mature as a computational process theory and has a much larger empirical and mathematical literature [Friston et al., 2017]. The present paper should not compete by claiming superior predictive machinery.

The distinction is instead one of derivation order and epistemic governance. Active inference generally begins with an agent model whose latent states, beliefs, preferences, and policies are already legitimate objects of the model. Readout Genesis places a prior methodological gate: semantic and agentic nouns are not admissible at the root merely because a modeller knows them in advance. The model must declare the translation by which such roles become accessible, the distinctions lost under that translation, and the claim strength permitted by the result. Formally, the difference is not "our dynamics are richer than active inference". It is

$$\begin{aligned}&\text{root retention} \not\equiv \text{belief state} \not\equiv \text{meaning}\\ &\not\equiv \text{experience} \not\equiv \text{choice}.\end{aligned} \tag{42}$$

A future empirical implementation may well instantiate parts of the human-domain dynamics using active-inference variables. Readout-native discipline would then treat those variables as one candidate domain realisation rather than as proof that the root was intrinsically Bayesian or semantic.

$$\delta_R \to S_n = (G_n, \Lambda_n, T_n) \xrightarrow{F} S_{n+1} \xrightarrow{q_D \text{ or } \tilde{q}_D} Z^{\text{cand}}_{D,n} \xrightarrow[\varepsilon_D]{\text{sufficiency / weld / readout gates}} D_n$$

$$\to x_n \xrightarrow{R_H} \rho^H_n \xrightarrow{\Psi_H} \mu_n \xrightarrow{\Phi_E} E_n \xrightarrow{\text{Retain}/U_H} H_{n+1} \xrightarrow{\text{history-shaped access}} \lambda^{\text{live}}_{A,n+1} \to L_{A,n+1}$$

$$\to \Pi^{\text{live}}_{A,n+1} \to \pi^{\text{choice}}_{A,n+1} \xrightarrow{X,F} \pi^{\text{act}}_{A,n+1} \to \text{world consequence / returned record} \xrightarrow{\text{Retain}} \delta_{R,n+1}$$

$$\text{Human–AI insertion:} \quad H_t \xrightarrow{L_H} Q_t \to \text{AI}_t \to Y_t \xrightarrow{R_H} E^{\text{AI}}_{H,t} \xrightarrow{\text{Retain}} H_{t+1}$$

$$\text{Session boundary:} \quad \text{AI removed} \to H_{\text{return}} \to \{\text{gain, loss, regulation, provenance, warrant, direction}\}.$$

Figure 1: Root-to-Human Master Equation River. The architecture is recursive rather than circular because retained change can alter the next reader. The full chain is not a single empirical law; it is a typed dependency architecture whose local components carry different epistemic statuses (Table 2).

15 Why This Is Not Simply Affordance or Capability Theory

Affordance theory and the capability approach already establish that real action possibility is relational and cannot be read from formal option lists alone [Gibson, 1979, Sen, 1985]. The Live Possibility Field earns a distinct role only if it captures a residual difference:

$$\text{feasible for the agent} \neq \text{currently live for the agent}. \tag{43}$$

A route may be legally open, materially feasible, and learnable within the horizon, yet fail to enter practical consideration because it is unnamed, unintelligible, socially unrecognised, affectively blocked, or absent from the current problem representation. Conversely, a route can be extremely live while remaining physically impossible or normatively dangerous.

This distinction generates a direct empirical burden. Under matched formal options and resources, a live-field measure should predict which non-cued alternatives people spontaneously generate, consider, reject, or return to later. If established affordance, capability, preference, salience, and cost measures explain the same variance, the live field should be compressed into those constructs rather than defended by terminology.

16 Why Human Return Is More Than Post-Test Performance

Human–AI studies often evaluate the coupled system while assistance is present, yet the programme's normative target concerns retained human capability. CTSA is not justified simply because delayed testing is sensible. It must add a useful classification of what returned and what was lost [Lahtee, 2026b].

The return architecture predicts partial dissociations. A user can acquire a new Concept without Skill; a Tool-selection rule without Alternative-generation; a Skill with poor metacognitive regulation; or alternatives that remain unwarranted. It also predicts gain–loss asymmetry: gross increase in one return dimension can coexist with reduced uncertainty sensitivity or checking habit in another. These are not empirical facts until tested. They are the minimal patterns that would make the Human Return interface explanatory rather than rhetorical.

17 Adversarial Tests and Rival Models

A theory programme becomes scientifically useful only when it can lose. The next empirical programme should not attempt to validate the entire river. It should test the smallest interfaces where the readout-native architecture claims incremental value.

17.1 Rival-model ladder

For a pre-choice study, compare at least

$$M_0 = \text{formal option count + stated preference}, \tag{44}$$
$$M_1 = \text{affordance/capability + cost/resources}, \tag{45}$$
$$M_2 = \text{constructed-preference / salience model}, \tag{46}$$
$$M_3 = \text{predictive or active-inference implementation}, \tag{47}$$
$$M_4 = \text{readout-native live-field model}. \tag{48}$$

The primary test is not in-sample fit. It is held-out prediction of spontaneous candidate generation, considered options, enacted choice, and delayed unaided re-entry. For Human–AI studies, hold current prompt content as constant as possible and vary interaction history, resistance route, or retrieval support. The pre-prompt claim earns a role only if prior interaction predicts later human judgements or candidate accessibility beyond current prompt, baseline traits, and simple exposure count.

17.2 Minimal empirical predictions

H1 — Live-field incremental value [OPEN]
Under matched feasible sets, $L_{A,t}$ will explain reproducible variance in which alternatives enter spontaneous practical consideration beyond resources, preference ratings, and ordinary salience measures.

H2 — History residual [OPEN]
Prior traversal history will predict later route accessibility after current semantic state, task, and control variables are matched. Failure removes the distinct momentum term.

Table 2: Claim-status ledger for the load-bearing transitions.

Object / transition	Status	Primary programme source	What would force revision or removal?
Retained distinction; ordered tape; finite root stepper	Root theorem / root architecture	Readout Universe; Readout Genesis [Lahtee, 2026m,l,k]	Fails only by revision of the root specification itself; does not imply human empirical validity.
Domain weld and no-silent-loss	Root architecture / protocol	Readout Genesis [Lahtee, 2026l]	Human-domain claims cannot inherit root status if translation defects or missing sufficiency remain unresolved.
Meaning after readout	Domain definition / architecture	MEMK [Readout Genesis Programme, 2026]	Remove any claim that treats meaning as root content; revise if a different domain translation explains the same target with fewer commitments.
Experience as meaning-giving	DR conceptual thesis	Experience Is Meaning-Giving [Lahtee, 2026c]	Narrow if significance-free lived experience can be operationally demonstrated under the declared scope.
History-shaped semantic access	OPEN empirical hypothesis	From Problem to Hypothesis [Lahtee, 2026j]	Remove the history term if current state, salience, task, and control explain later access without a residual path-history effect.
Resonance / rhythm	OPEN optional domain diagnostics	Experience Is Meaning-Giving [Lahtee, 2026c]	Drop if schema/familiarity/retrieval or ordinary timing variables explain the same phenomena with no incremental value.
Live Possibility Field	DR / OPEN measurement object	Choice Begins Before Choice [Lahtee, 2026a]	Compress into capability/affordance/preference measures if it adds no reliable information about spontaneous candidate availability or delayed re-entry.
Effective agency envelope (witnessed)	Measurement architecture; definitions at DEFINITION; lemmas on a stated discrete model	Potential as a Readout [Lahtee, 2026i]	Revise if read/decision/enactment/feedback separation does not improve identification over simpler outcome measures in declared tasks.
Candidate status and D-R-A	Domain rule / constitutive necessity under the target definition	Fusion/Tunnel [Lahtee, 2026d]	D-R-A changes if the target definition of durable human epistemic expansion changes; none is a truth guarantee.
CTSA Human Return	OPEN measurement architecture	CTSA v9.1 [Lahtee, 2026b]	Narrow or remove if classes are unreliable or add no information beyond ordinary delayed performance, transfer, and established learning measures.

H3 — Pre-prompt carryover [externally constrained, mechanism OPEN]
Repeated Human–AI interaction can alter later human judgements before a new AI response is seen. External evidence supports the general carryover phenomenon [Glickman and Sharot, 2025]; the readout-native mechanism remains open.

H4 — Human Return dissociation [OPEN]
Delayed unaided Concept, Tool, Skill, and Alternative Return will show partially dissociable profiles rather than collapse into one generic performance factor.

H5 — Gain/loss asymmetry [OPEN]
Positive gross return can coexist with reduced checking habits, uncertainty sensitivity, or alternative accessibility; net human expansion will therefore differ from gross retained gain.

H6 — Optional modifiers must earn survival [OPEN]
Resonance, rhythm, and barrier/transition variables remain only if they add held-out explanatory or predictive value beyond schema, familiarity, appraisal, retrieval, exposure count, event boundaries, and continuous accumulation models.

18 GRACEFUL DEGRADATION

The programme should survive the deletion of its most provocative metaphors. If Human LoRA terminology is removed, the remaining claim is selective retained operator change. If resonance is removed, current-versus-retained congruence can be represented by ordinary schema/retrieval variables. If rhythm is removed, ordered-history and event-boundary variables remain. If barrier language is removed, continuous retained accumulation remains available. If CTSA does not outperform simpler delayed measures, Human Return can be retained without the four-class taxonomy. The load-bearing core is narrower:

$$\begin{aligned}&\text{retained distinction} \to \text{domain translation/readout}\\&\to \text{meaning} \to \text{experience} \to \text{retained change}\\&\to \text{graded accessibility} \to \text{live possibility}\\&\to \text{choice/action} \to \text{feedback/new record},\end{aligned} \tag{49}$$

with provenance, loss, reader, claim status, and correction kept explicit. If this core itself adds no explanatory or methodological value beyond existing architectures, the programme should be reduced to a vocabulary for organising already known distinctions. The architecture earns theory status only through discriminating use.

19 Contributions

The paper's contribution can now be stated without claiming ownership over its mature neighbours.

A root-to-human derivation order. Meaning, experience, live possibility, choice, agency, and Human–AI return are not inserted at the root but admitted downstream of retained difference and translation.

Domain weld and defect accounting as epistemic governance. Structure-preserving abstraction is established mathematics; the contribution is to use it as a guard against semantic import and certainty drift.

A continuous but non-collapsing chain from meaning to agency. Experience is downstream of readout; live possibility is downstream of meaning and history but upstream of choice; agency is broader than observed action.

Human–AI interaction inside the same recursive river. A prompt is generated from a pre-existing human state, AI output begins as a candidate, and retained effects can reshape the next human state before the next prompt.

Human Return as the upper measurement boundary. Assisted performance and durable human capability are intentionally separated.

A claim-status and provenance ledger attached to the equations themselves. A transition is incomplete unless its reader, source, defects, claim tier, and legitimate defeater are visible.

20 Conclusion: Before Meaning, Before Choice

The strongest insight of the Human–AI Readout Programme is not that experience contains meaning or that choice has social and historical conditions. Those propositions have deep predecessors. The stronger contribution is methodological: do not begin where familiar human vocabulary makes the problem look easiest.

Begin with what is retained and distinguishable. Ask what reader makes which structure accessible. Require a domain translation before semantic naming. Expose what that translation loses. Treat meaning as a domain significance relation rather than root substance. Treat experience as the lived configuration produced when an encountered trace becomes significant to a finite reader. Treat retention as selective rather than automatic. Treat accessibility as graded and history-shaped. Treat live possibility as distinct from feasible capacity. Treat choice as downstream of what has become live. Treat action, outcome, and evaluator-visible record as different levels. Treat AI output as a candidate encounter rather than truth. Treat retained human change as direction-neutral until resistance and warrant are checked. Finally, remove the AI and ask what came back with the human.

The resulting thesis is therefore not a new psychological slogan but a derivational constraint:

$$\begin{aligned}&\text{Retained Difference} \to \text{Admissible Readout}\\&\to \text{Meaning} \to \text{Experience}\\&\to \text{History-Shaped Accessibility} \to \text{Live Possibility}\\&\to \text{Choice} \to \text{Corrigible Agency}\\&\to \text{World Consequence} \to \text{New Retained Difference}.\end{aligned} \tag{50}$$

Human–AI systems enter this river as mediating structures, not as epistemic sovereigns. Their importance lies in how they change what can be articulated, compared, retained, resisted, and returned. The next scientific step is consequently modest but decisive: test whether the readout-native interfaces predict anything that simpler rival models do not. If they do, the programme has identified a useful architecture. If they do not, the architecture must shrink.

21 Source Lineage and Check Record

Version 1.0 of this paper was checked with the glosa methodology [Lahtee, 2026e] before deposit. An independent review pass found no defect in the argument, the notation, or the attributions to the deposited programme papers, and found the external attributions it could check at full text (Tishby et al.; Glickman and Sharot) accurate; it found three source-lineage defects, all repaired here. (i) The Readout Genesis whitepaper was cited as a floating "working specification"; it is a public file on the main branch of `morrocwi/readout_genesis` and is now cited by commit and blob SHA. (ii) The MEMK domain specification was cited as an unpublished "v0.3.0 ANCHORED" draft; the only MEMK file in a public tree is v0.2.0 on the branch `feat/register-philosophy-memk-v0.2.0`, cited here by commit, blob, and SHA-256, with the v0.3.0 draft disclosed as unpublished and nothing in the paper depending on it. (iii) The agency-potential model was cited from a superseded, undeposited programme note and written over Π^{feas}; it is now cited from *Potential as a Readout* and written over the witnessed set (Section 9). Every other internal reference now carries its Zenodo DOI, and *Master Equation River* is cited in its glosa-checked v1.1. The three git-anchored citation cards and the cards for the world literature named in Section 2 live in the glosa repository (`records/lit/before-meaning-before-choice`), each carrying the fetched URL, page, line, and one verbatim passage, re-located by a fresh headless session of the same model (independence class I1). Knowledge state: K0 for every claim, timestamped and citable, not yet checked at I2 or above.

AI-assistance disclosure. The rebuild of v1.0 into this version, the source fetching, the git anchoring, and the passage extraction were done with an AI assistant under the author's direction; the problem, the argument, the selection of what to claim, and the v1.0 text that anchors this version are the author's. No AI or vendor is an author. The author's raw lines behind this version are in the Blackbox Log (concept DOI 10.5281/zenodo.22302518, entries BBL-2026-09-06-144 to -149).

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